

Aishwarya Soni

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EDUCATION

• Indian Institute of Technology, Kharagpur

2018 – 2023

- *M. Tech Dual Degree (B.Tech + M.Tech), **Electronics & Electrical Communication Engg.**; CGPA: 8.6 / 10*
- Minor: **Computer Science & Engineering** | CGPA: 8.75 / 10
- Micro-specialisation: **Embedded Control, Software, Modelling & Design**

EXPERIENCE

Honeywell Connected Enterprise (HCE)

Bengaluru, India

Software Development Engineer 2

Jul 2023 – Present

Forge Operation Center (FOC):

- Built a full-stack **React + Django + PostgreSQL** dashboard with multiple pages tailored for product owners, vertical heads, and C-suite — used in monthly accountability reviews, investor presentations, and market expansion drives
- Co-designed a unified PostgreSQL schema to normalise and consolidate heterogeneous data from **30** product sources — enabling consistent KPI, churn, and **site-health** reporting across teams with vastly different underlying data models
- Engineered multi-source data ingestion using **Azure Service Bus, IoT Hub, Kafka**, and scheduled **Celery** workers on **AKS** — aggregating KPI and performance metrics across **25K** customers, **282K** sites, and **5M** assets
- Implemented a **Redis cache-aside** layer for dashboard API endpoints — selectively caching and invalidating relevant keys with cache-miss refill logic, reducing dashboard API response times from 30–60 sec to under **2 sec**
- Built platform reliability infrastructure using **ELK** for log aggregation and automated alerting on pipeline failures — with 5-minute cron-based health checks ensuring continuous monitoring across **40K** gateways and all ingestion workflows

Platform Health Dashboard:

- Designed a fully config-driven architecture where services, data sources, thresholds, and alert rules are managed via **Django admin** and **Azure Key Vault** — enabling zero-code onboarding of new services with no engineering changes across **50** services and 8 modules
- Architected a 3-tier time-series storage strategy — per-minute snapshots with 1-day retention, hourly and daily aggregates in separate tables — powering auto-granular trend views and real-time threshold breach detection across all monitored services
- Built a source-agnostic health ingestion layer with custom adapters for **InfluxDB, Azure IoT Hub, Databricks**, and Honeywell IoT health-check APIs — with a 5-minute **Celery** cron evaluating configurable thresholds and dispatching targeted email alerts
- Built an **agentic OpenAI** RCA pipeline that traverses service dependency graphs and retrieves historical RCA annotations to execute multi-step root-cause and downstream impact analysis for on-call SRE teams

SaaSops Portal:

- Built a centralised **Azure** service-principal secret management service with **RBAC**-gated access, 30-day expiry alerting, naming conventions, and auto-propagation to **Octopus** across 5 tenants and 100+ principals
- Developed a **React + Django** operations dashboard within a **microservices** architecture, tracking ~1,000 tenant jobs with business-group filters, real-time status visibility, and failure monitoring across **Databricks** pipelines

SKILLS

- **Languages:** Python, JavaScript / TypeScript, C / C++, SQL
- **Backend & Frameworks:** Django, Node.js, Celery, REST APIs, GraphQL
- **Data & Messaging:** Kafka, Redis, PostgreSQL, MySQL, InfluxDB, Victoria Metrics, Databricks
- **Cloud & Tools:** Azure (IoT Hub, Event Grid, Service Bus), Git, OpenAI API, Google Earth / CesiumJS
- **Frontend:** React, HTML5 / CSS3

AWARDS & ACHIEVEMENTS

National Standard Examination in Astronomy (NSEA) — State Rank 2 in Gujarat **2017**
JEE Advanced — All India Rank 1050 among 1.5 million candidates (Top 0.07%) **2018**
Inter-IIT General Championship — **Gold Medal**, Fine Arts Cup (1st among 22 IITs) **2021**
Inter-IIT General Championship — **Silver Medal**, Fine Arts Cup; **Silver Medal**, Music Cup **2022**
Inter-IIT General Championship — **Gold Medal**, Music Cup (1st among all IITs nationwide) **2023**
IEEE ISIT — Published “*Bayesian Persuasion: From Persuasion toward Counter-suasion*” (2nd author) **2024**
Honeywell Performance Excellence Award — **6× Silver** and **1× Bronze** for product & platform engineering... **2025–26**

